

Sofia L. Cardona

+1 (612) 281-6602 | scardon2@nd.edu | San José, Costa Rica | [LinkedIn](#)

EDUCATION

University of Notre Dame – College of Engineering – Bachelor of Science in Engineering
Major: Computer Science | **Minor:** Engineering Corporate Practice
Engineering Summer Abroad 2024, Science and Engineering of Digital Photography and Ethics
Coursework: Data Structures (C/Python), Fundamentals of Computing (C), Systems Programming, Discrete Mathematics

RELATED COURSEWORK OR PROJECTS

Incoming Software Engineer Intern | Norwalk, Connecticut **May 2025**

FactSet

- Selected for FactSet's MESH (Mentor, Support, Engage, Hire) software engineering internship focused on fintech.

Engineering Computing Application | University of Notre Dame **March - April 2024**

- Collaborated on coding an application on MATLAB to analyze and visualize global methane emissions data, detailing emissions by continent, country, and specific contributors.

Wordle, Game | University of Notre Dame **August - December 2023**

- Developed a Wordle-inspired game in C, utilizing arrays and visual elements for an interactive experience.

Permeate Probe Data Analysis – Research Assistant | University of Notre Dame **January 2024 – Present**

Dowling Lab

- Developed and implemented computational models in Python using Pyomo, IDAES, and NumPy to analyze permeate probe data from a high-throughput diafiltration apparatus, optimizing membrane transport parameter estimation.
- Utilized advanced statistical and computational methods to identify transport mechanisms, contributing to parameter estimation and model calibration for membrane separations.

Student International Business Council (Strategy and Technology Consulting)

PwC, Project Leader | Chicago, Illinois **February 2025 - Present**

- Leading a team of 8 students in a strategic consulting engagement, developing a cybersecurity risk mitigation strategy for a Fortune 500 e-commerce client, analyzing a major data breach and identifying key vulnerabilities.
- Developing a strategic, data-driven response for executives, focusing on crisis response, incident containment, risk mitigation, and cybersecurity governance to minimize financial and reputational risk.

Bosch, Travel Team Analyst | Chicago, Illinois **February - April 2024**

- Analyzed and proposed innovative strategies for integrating the Metaverse into Bosch's recruitment processes, focusing on enhancing user engagement and candidate experience.
- Presented actionable recommendations to Bosch's HR and Tech teams for potential implementation.

Amazon, Travel Team Analyst | Boston, Massachusetts **October – November 2024**

- Partnered with a team to redesign Amazon's main page interface using Figma, optimizing underutilized features to improve user engagement functionality.
- Incorporated Social Media and Generative AI elements to drive innovation and enhance the user experience.

LEADERSHIP AND ACTIVITIES

Treasurer, Computer Club at Notre Dame | University of Notre Dame **October 2024 - Present**

- Leading efforts to restart the Computer Club in collaboration with graduate students, focusing on re-building a strong computer science community.
- Organized and managed events, workshops, and networking sessions to enhance members' technical skills, foster collaboration, and expand industry connections.

Society of Women Engineers Member | University of Notre Dame **February 2024 - Present**

- Attended the *SWE National Conference*, gaining recruiting insights and networking with engineering professionals. Participate in professional development events to expand industry knowledge.

EXPERIENCE

Marketing for Tauro | University of Notre Dame **February - May 2024**

- Contributed to the creation and promotion of a student-led restaurant startup, advancing it to the McCloskey competition. Designed and executed digital and social media strategies to enhance brand visibility and engagement.

Academic Peer Mentor | University of Notre Dame **August 2024 - Present**

- Mentored a first-year Computer Science student from the Greater China Scholars program, offering guidance academic balance and support through periodic meetings and discussions.

TECHNICAL AND LANGUAGE SKILLS

Technical skills: C, C+, Python, Bash, MATLAB, Microsoft Excel, SolidWorks, 3D Printing

Language: English (proficient), Spanish (proficient)